



American International University-Bangladesh (AIUB)

Department of Computer Science Faculty of Science & Technology (FST)

“DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM”

A Software Engineering Project Submitted

By

Semester: Spring_24_25		Section: G	Group Number: 02	
SN	Student Name	Student ID	Contribution (CO3)	Individual Marks
	AKASH ROY	22-48054-2		
	ABDULLAH AL TAIEB	22-48028-2		
	MD.REZAUL ISLAM	22-48040-2		
	HASIBUL HASAN	22-48026-2		

The project will be evaluated for the following Course Outcomes

CO3: Select appropriate software engineering models, project management roles, and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation Regarding Process Model Selection	[5Marks]
	Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]
Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
CO4: Develop a project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
	Develop the project plan, its components of the proposed software products	[5Marks]
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	

Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	
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Description of Student's Contribution in the Project work

Student Name: AKASH ROY Student ID: 22-48054-2 Contribution in Percentage (%): 25% <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 _____ AKASH _____ Signature of the Student
Student Name: ABDULLAH AL TAIEB Student ID: 22-48028-2 Contribution in Percentage (%): 25% <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 _____ TAIEB _____ Signature of the Student
Student Name: MD. REZAUL ISLAM Student ID: 22-48040-2 Contribution in Percentage (%): 25% <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 _____ REZAUL _____ Signature of the Student
Student Name: HASIBUL HASAN Student ID: 22-48026-2 Contribution in Percentage (%): 25% <u>Contribution in the Project:</u>

- Contribution Description 1
- Contribution Description 2

____HASIBUL____
Signature of the Student

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing	Some errors in spelling and grammar. Some problems	Few errors in spelling and grammar. Presents most of the details in	Project report is complete and No errors in spelling and	

	organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	a logical flow of organization in definition, details, and example.	grammar. Consistently presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

1. PROJECT PROPOSAL

1.1 Background to the Problem

In Bangladesh, as a growing economy we are dealing with several social and economic challenges. Problems like inflation, corruption and scarcities are a great burden on the growth of our economy. Our economy mainly depends on agriculture. More than 250 million people are directly connected with it. Though we are dealing with a huge price hike, the fate of our farmers has not changed.

Traditionally, our supply chain is dominated by the middlemen. They often create artificial problems like scarcity and surplus. They earn a huge profit margin from both ends and discourage the farmers from producing food. Due to the lack of preservation facilities, farmers are forced to sell their products. Sometimes, to earn extra money they use prohibited preservatives. Which is highly concerning for our health. There is also a lack of documentation on the demand and supply chain of the food. It is almost impossible for individual rural farmers to survive in this big marketplace. These are places where technology can play a great role.

Root Cause of the problem:

- Middlemen take the most profit margin. Though consumers pay high prices the fate of the farmers is unchanged.

- Farmers are forced to sell their products at very cheap prices because of artificial surplus created by syndicate. They also create artificial shortage to sell at more price.
- The lack of transparency in using chemicals and pesticides.
- The medium or small cannot compete and survive in the domestic market due to syndicate.
- Lack of communications between the consumer and producer.
- Lack of availability of user-friendly services.

1.2 Solution to the Problem

Objective

Our objective is to create a direct farm-to-consumer marketplace by ensuring seamless and transparent transactions between farmers and consumers. The aim is to empower farmers by generating sustainable revenue and give them an opportunity to directly access the domestic marketplace by crossing traditional barriers. Where they will be able to sell their products directly to consumers, ensuring fair pricing and will offer extra profit margin. This approach will benefit the consumers also by providing fresh, high-quality products at reasonable prices. A key goal of this project is to break the syndicates and reduce the domination of mediokers in the agricultural market.

Solution:

Technology will come to a great use tackling the issues. We propose to create a web app named “Direct Farm to Consumer” which will eventually connect the producer and the customers. This will allow farmers to register and display their products, setting their own prices based on true production costs and so on. On the other hand, consumers can browse these products, make purchases, and get delivery directly from the farmers. By eliminating middlemen, this model will allow farmers to get a fair price for their produced goods while it also offers consumers extra money.

Additionally, it will increase transparency by allowing farmers to disclose their cultivation process. We aim to encourage organic and sustainable practices, ensuring consumers safe, high-quality, unadulterated products. This will also help farmers to step into the domestic markets and to compete alongside the giant companies. This will help to eliminate syndicates and help our economy. It will work as a connecting bridge between the rural producer to the urban consumers across the country.

Target group: The primary target users are small to medium-scale farmers and urban consumers.

For Farmers: They can gain direct access to domestic market, pricing control and full profit from sales without an intermediary and will be able to establish his own business without any extra costs.

For Consumers: They will be benefited from affordable, fresh, healthy goods with hassle free delivery to save their time and money.

By studying and documenting the economic impact of eliminating intermediaries, this platform will provide valuable data for researchers on rural economics models.

Literature Review:

- Saniya Firoj Kalawant , Alisha Sameer Mujawar , Dipak Rajaram Lohar, proposed a “E-Shop-Fruits and Vegetable Store”. The project offers an online shopping platform for fresh fruits and vegetables with convenience, time- saving, and fast delivery services.
- Ms. Shradha G. Gobare , Prof. Sachin M. Dandage , Dr. P. • M. Jawandhiya, introduced a “E-Mandi an E-Vegetable Market.” This project aims to enhance the vegetable market’s business and transparency by creating a website for consumers, retailers, sellers, and farmers.

In comparison with existing software solutions, this approach offers more user friendliness and flexibility. It connects consumers and farmers directly to each other from anywhere in the country. It eliminates the middlemen and helps to break syndicate. Though there are many software solutions, but none is introduced in Bangladesh that connects producer and consumer like this platform.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

Our project is mainly based on direct interaction between farmers and consumers using digital platforms. It is not possible to select all the functional requirements at once because the project which we are not going to use is a linear model. The linear model is not suitable because the completion of one phase depends on the completion of previous phases. Iterative process models are also not usable.

Though the phases can be reviewed and changed after the completion of one phase, it does not include customer involvement. So, our proposed methodology would be Agile Methodology. And the model would be scrum.

Scrum is an agile methodology using an approach that is both iterative and incremental. There are frequent assessments of the project, which enhances productivity and reduces the risk of failure. Scrum is based on three pillars: transparency, inspection, and adaptation. Scrum encourages frequent inspection on progress of the project. The inspections, however, do not happen so frequently to hamper development. Developers sometimes are hesitant to use scrum because it points out most of the things they are not doing right. When any issue is detected, the team must act to correct the problem.

Reasons for choosing scrum:

1. Scrum provides short iterations and fast feature Delivery. Scrum takes usually 1–4 weeks to deliver features. Our platform needs rapid deployment of new features like sales, offers, recommendations etc.
2. The incremental approach makes developing complex features easier.
3. Scrum takes feedback at the end of each sprint from user reviews. It goes perfectly with our customer centric approach.
4. Teams stay connected through daily meetings, which is essential for a fast development approach. It helps detect issues early and saves time. There could also be multiple stakeholders.
5. Scrum allows us to change even late in development. So, based on market trends and customer behavior flexibility offered by scrum is necessary.

Scrum provides flexibility, feedback, stakeholder engagement and speed which are necessary for building a Direct-Farm-to-Consumer platform.

2.2 Project Role Identification and Responsibilities

Roles and Stakeholders in Software/Project Management Activities:

1. **Product Owner:** Acts as the main communicator between the development team and stakeholders.
2. **Scrum Master:** Stays connected with the owner and ensures that the team follows scrum practices.
3. **Scrum Development Team:** Designs, codes, tests, and deploys features during each sprint.
4. **Stakeholders [Farmers, consumer, Admins]:** Actively provide feedback on features and ensure their expectations are met.
5. **UX/UI Designer:** Designs user-friendly interfaces.
6. **Quality Assurance (QA) Tester:** Tests each sprint's deliverables and ensures the fault proneness to maintain quality.

3. REQUIREMENT ANALYSIS

Global Functionalities:

Sign Up:

Functional Requirements:

- Users need to sign up with their username, phone number/email and password (Weak passwords will be rejected.)
- The sign-up credentials will be verified with OTP or confirmed through email/message.
- The system should check duplicate registrations using phone numbers or email.
- There will be a checkbox if the user wants to sign up as a farmer/consumer/admin.
- It will take their name as input and save all their input in database with proper security
- There will be a link if the user has already signed up and if he wants to sign in.

Non-Functional requirements:

- There will be another sign-up option to sign up through google or Facebook API.
- It must ensure secure data storage for user security using encryption [Using JWT web token, Auth 2.0].
- The system needs to handle many user registrations and handle crashes.

Development Constraints:

- The systems must be compatible with all kinds of data protection laws.
- The backend should be scalable to support increasing user registrations.

Priority Level: High

Log In:

Functional requirements:

- The software will allow users to sign in their username (email/mobile number) or password.
- It will also allow users to sign in with google or Facebook.
- There will be password recovery options in case user forget the password.
- It will authenticate the user by checking the provided username or password matched with the username and password of database.
- If a user tries more than three times with the wrong password it will restrict any new sign in request within the next 30 minutes.
- There will be a link if the user is not signed up and if he wants to sign up.

Non-Functional Requirements

- The system should complete authentication within 2 seconds under normal conditions.
- The system should ensure around 95% to 99.9% up time to provide high availability.

Development Constraints

- The system must be integrated with Google and Facebook authentication
- The system should display generic error messages like “Invalid credentials” to prevent exposing sensitive information.

Priority Level: High

Nav Bar Setup:**Functional Requirements:**

- There is a "Fresh Everyday" logo at the top.
- A circular placeholder for the user's profile image.

Profile Page Setup:**Functional Requirements:**

- There will be a Change Picture Button/Delete picture Button.
- Users can change the previous name, phone/email from the desired placeholder.
- User can address in the address box. Users will be able to add delivery/business address based on their need. [Except admin]
- A button to allow users to save all profile information.

Non-Functional Requirements:

- The system should complete the update within 2 seconds.
- The profile setup feature should be accessible 24/7.

Development Constraints:

- Profile images should be stored using cloud storage solutions and limited to 5MB per file.
- The system should use SMS or an email verification service for contact number changes.

Priority Level: Moderate

Individual functionalities:

Farmer Dashboard

Functional Requirements:

- There is a navigation menu on the left side. In this navigation menu, there are multiple buttons like (Product list, add product, History, sales, messages).
- By clicking the product button, the user can see the product list. In this product list, users can view all available products.
- Add-new-product button allows users to add new products for sale. A new window will open to upload name, photo, price, description.
- If the user clicks the order button, the user can view current and past order history.
- If a user clicks this sales button, the user can check today's sales records.
- Messages button allows users to send and receive messages from customers or support.
- There will be also a category of options to filter the product as its price and quality.
- Today's Sell box shows the total sales amount for today and there will be another box like today's earnings box. This box shows users' total earnings for day, week, month and year.
- There is a section that shows a graphical overview of last month's sales performance.
- There is also available a panel that displays the top-selling items.

Non-functional Requirements:

- Search and filtering should be fast and efficient.
- The user interface should be simple and accessible.

Development Constraints:

- Deployed on cloud platforms like AWS or Firebase.

- Must be cost-effective and scalable.

Priority Level: High

Consumer Dashboard Page:

Functional Requirements:

- Home button allows users to go to the homepage where user can see featured products.
- Favorites button allows users to see my favorite products that user have saved for later.
- Cart button allows users to see the products that user have added to the shopping cart.
- Order button allows users to check their previous and current orders.
- Track-Order button allows users to track their product deliveries and see their status.
- Help & Information button allows users to get support.
- Log Out button allows users to log out from the account.
- There will be a notifications icon. User can see the todays or previous all of product or system notification like ("You got a 20% off promo", "Your product has reached the warehouse.", "Your order is confirmed.")

Non-Functional Requirements:

- The dashboard shall have a user-friendly and responsive UI.
- Pages and notifications shall load within 2-5 seconds.
- It should handle high traffic loads and be compatible with mobile, tablet, and desktop.

Development Constraints:

- The system shall support API integrations for order tracking, payments, and notifications.
- The development timeline shall be 3-6 months.

Priority Level: High

Admin Dashboard:

Functional Requirements:

- Consumer/Farmer Management button allows admin to manage consumer accounts and information.
- Manage Orders button allows admin to view and manage all system orders
- Reports and Feedback button allows admin to access system reports and user feedback

- Logout button allows admin to sign out of the admin account
- Total Consumer/Farmer placeholder shows the total number of registered consumers.
- Total Orders placeholder shows the total number of orders processed through the system.
- There is a bar graph showing monthly sales data and user growth

Non-Functional Requirements:

- The interface must be user-friendly with smooth navigation.
- Security measures like authentication and data encryption must be in place.
- The system should ensure 99.9% uptime with daily backups.

Development Constraints:

- The database should be MongoDB or MySQL.
- The system should be hosted on a cloud-based platform like AWS or Firebase

Priority Level: High

Consumer Home Page:

Functional Requirements:

- There is a search bar. Consumers search in this search bar to find products.
- Shopping cart icon. If consumers purchase a product, this product goes to the cart.
- "Become a Farmer" button with farmer icon for as a farmer registration.
- Language toggle button for Bengali.
- Home button to view the main product page.
- Categories button to filter products by category.
- Discount Product images allow us to see special offers and discounts.
- Farmers Near You button to locate local farmers.
- Help button to access support.
- Product images, names, price and description.
- The comment button allows users to comment on specific products.

- Add to Cart button

Non-Functional Requirements

- The website should be user-friendly and responsive across devices.
- The system should support growing users and products efficiently.

Development Constraints

- The project should be built using modern web technologies like React, NextJs, MongoDB, ExpressJs and Node.js.
- Development should be completed within six months.

Priority Level: High

Payment Page:

Functional Requirements:

- There is a title “PAYMENT” at the top of the page.
- bKash payment option with logo and Tap button.
- Nagad payment option with logo and Tap button.
- Mobile Number and Pin Code input field.
- Cancel button to abort the payment process.
- Pay button to complete the transaction.

Non-Functional Requirements:

- The system must provide clear success or failure messages.
- Payment processing should be fast and reliable.

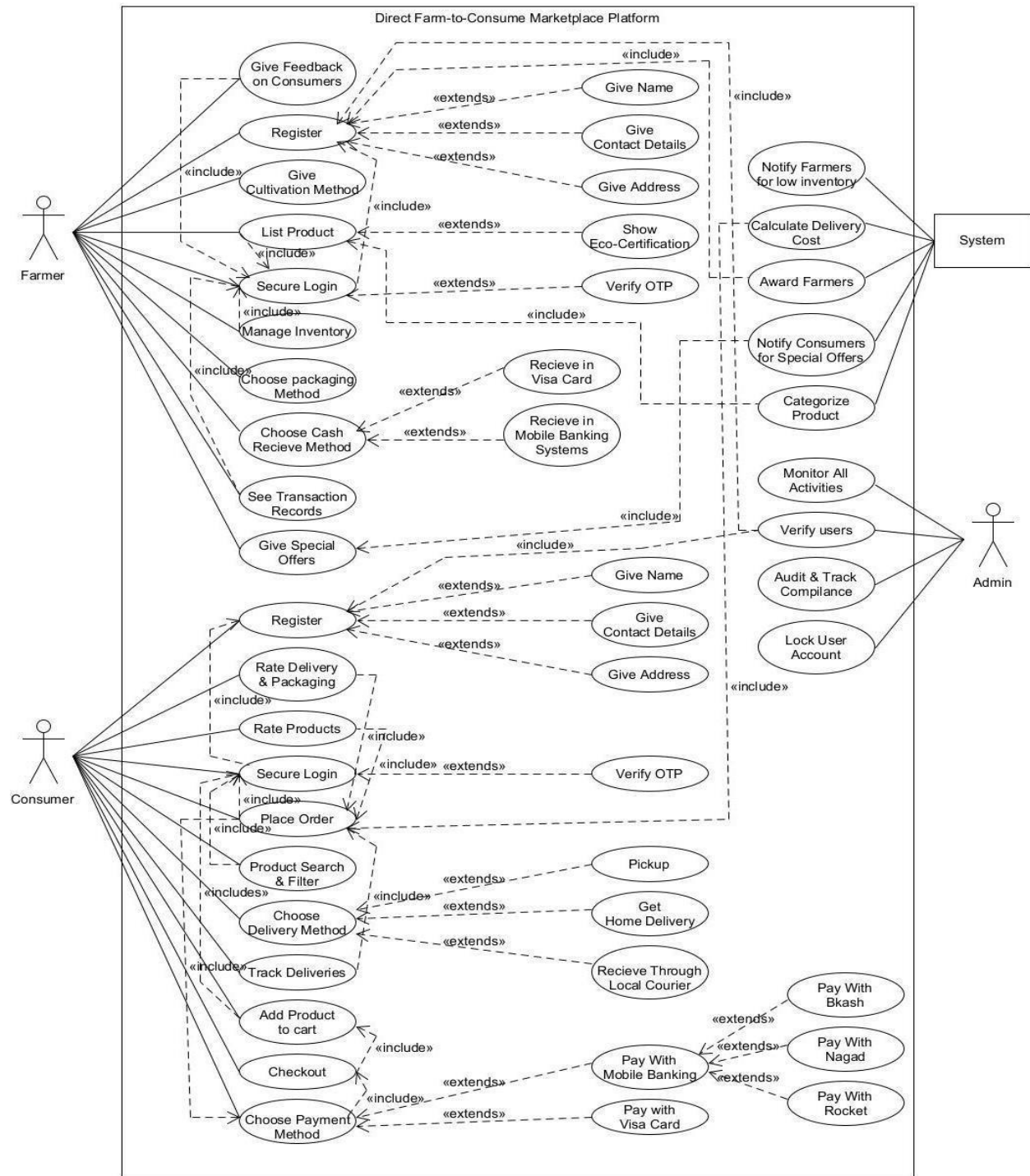
Development Constraints:

- The system must follow local financial regulations.
- It must integrate with official payment APIs.

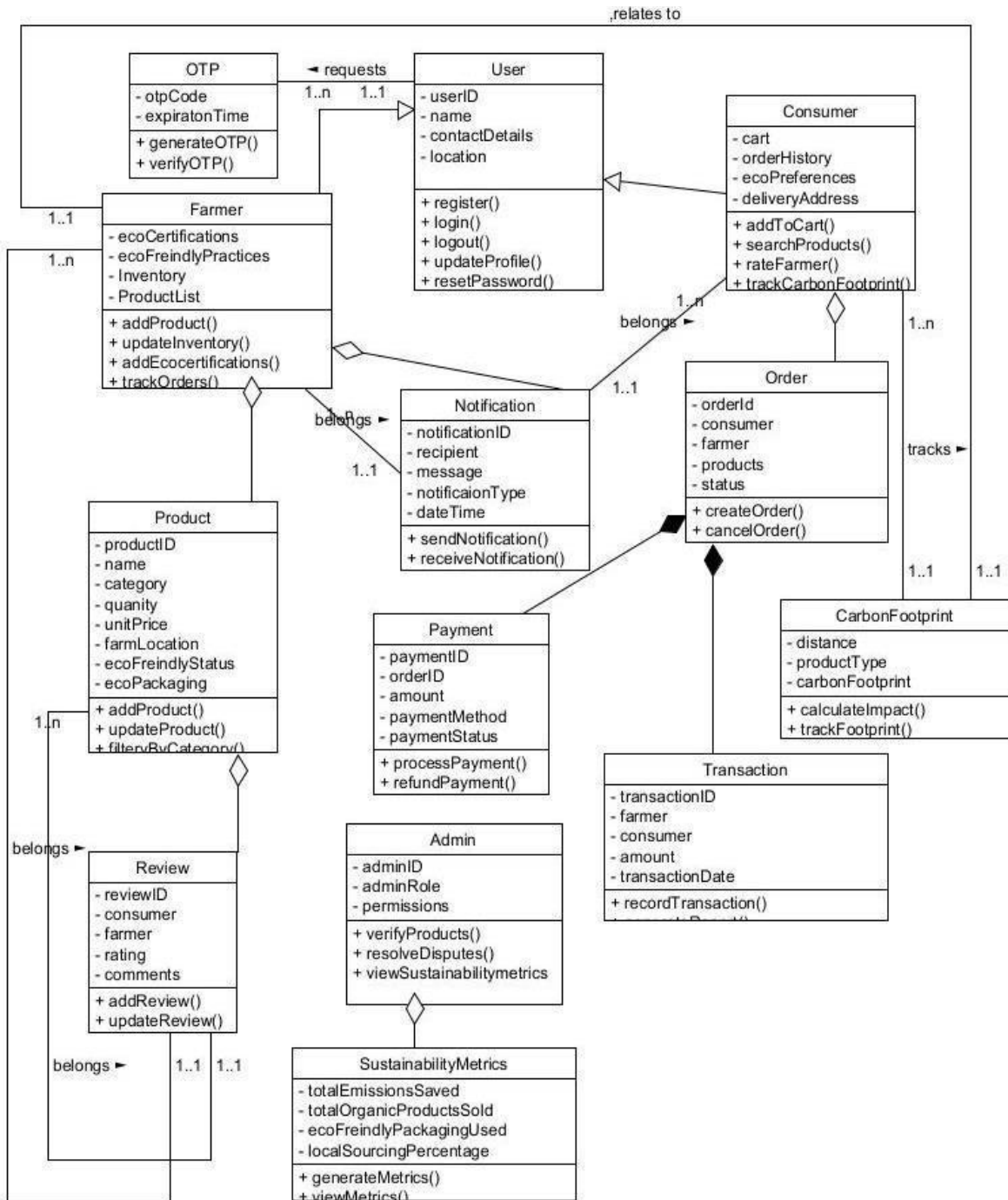
Priority Level: Moderate

4. DIAGRAM

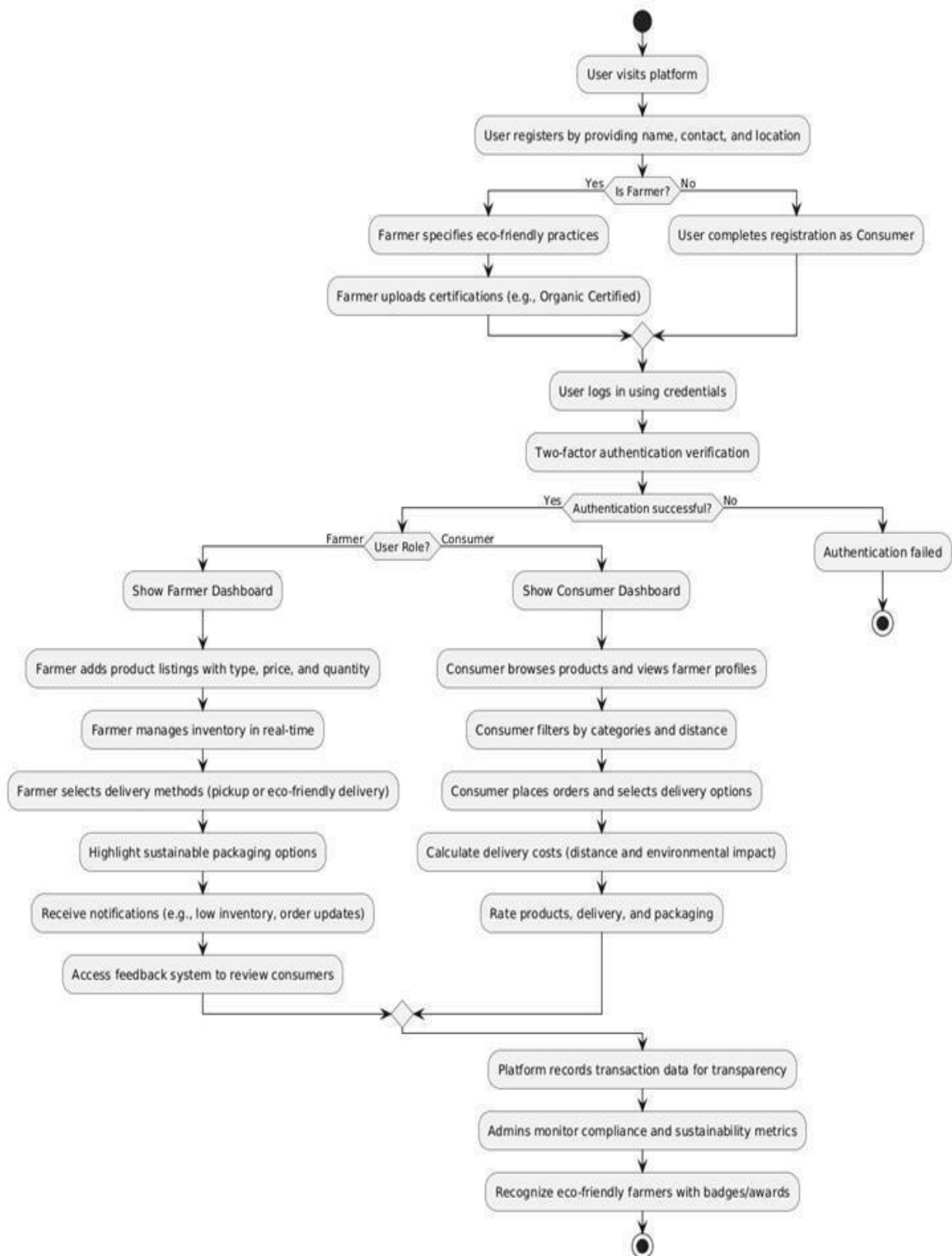
USE CASE DIAGRAM



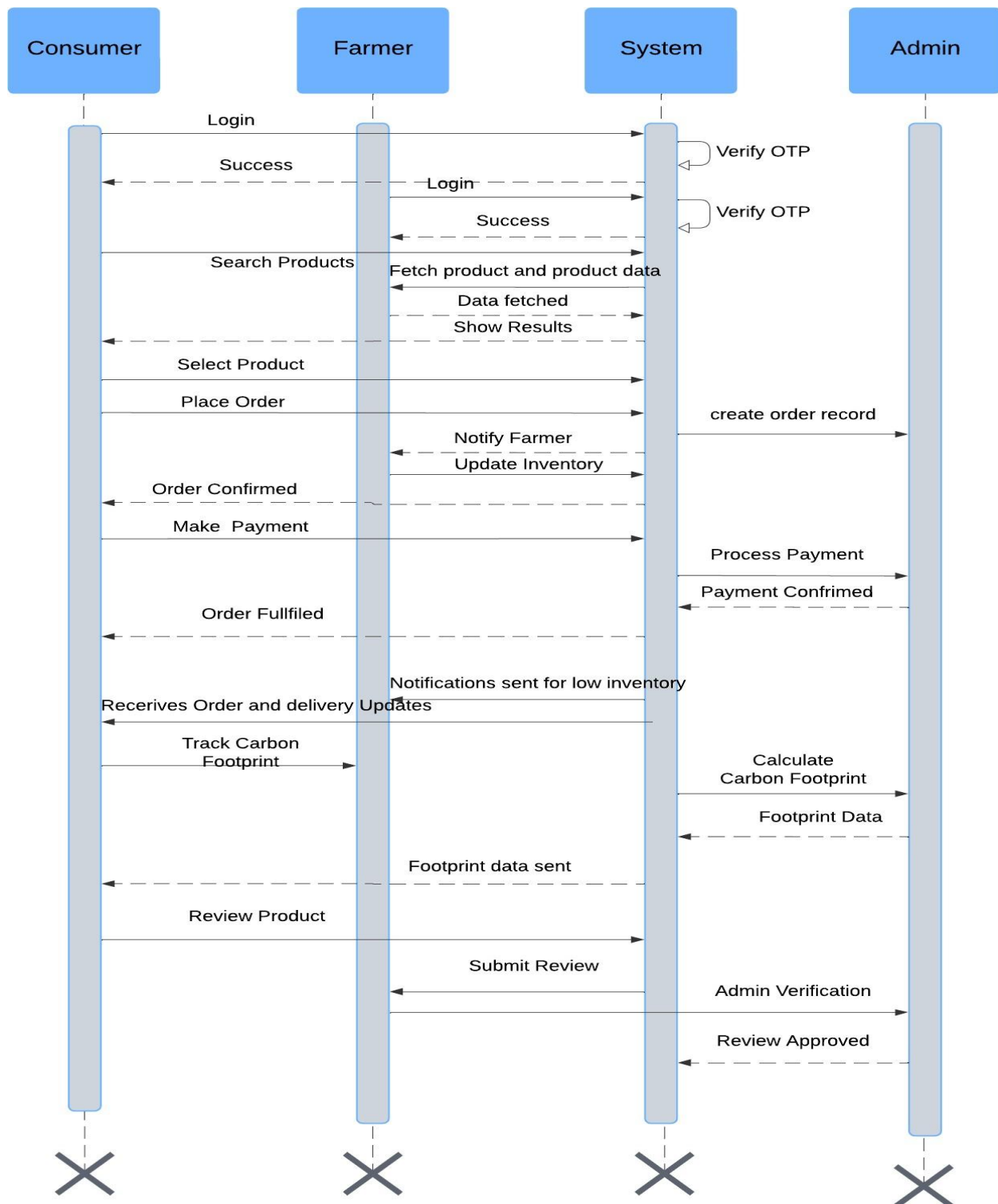
CLASS DIAGRAM



ACTIVITY DIAGRAM



SEQUENCE DIAGRAM



Rubric for Project Assessment (CO4)

Marks Distribution (Maximum 3X5=15)					
Marking Criteria	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	Acquired Marks
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					

1. UI UX Design

Fresh Everyday



Welcome Back

-Get Connected With-



Create new Account

Already Registered? Login

Please enter your name

Please enter Email or Phone number

Please enter password

Please enter date of birth

Please enter your Full Address

SIGN UP

Fig-Sign in Page

Fresh Everyday



Log in using social media to get quick access

 Signing With Facebook
 Signing With Google

Login

Sign in to continue

Please enter Email or Phone number

Please enter password



Forgot Password ?

LOGIN

Don't have an account? Signup here

Fig-Log in Page

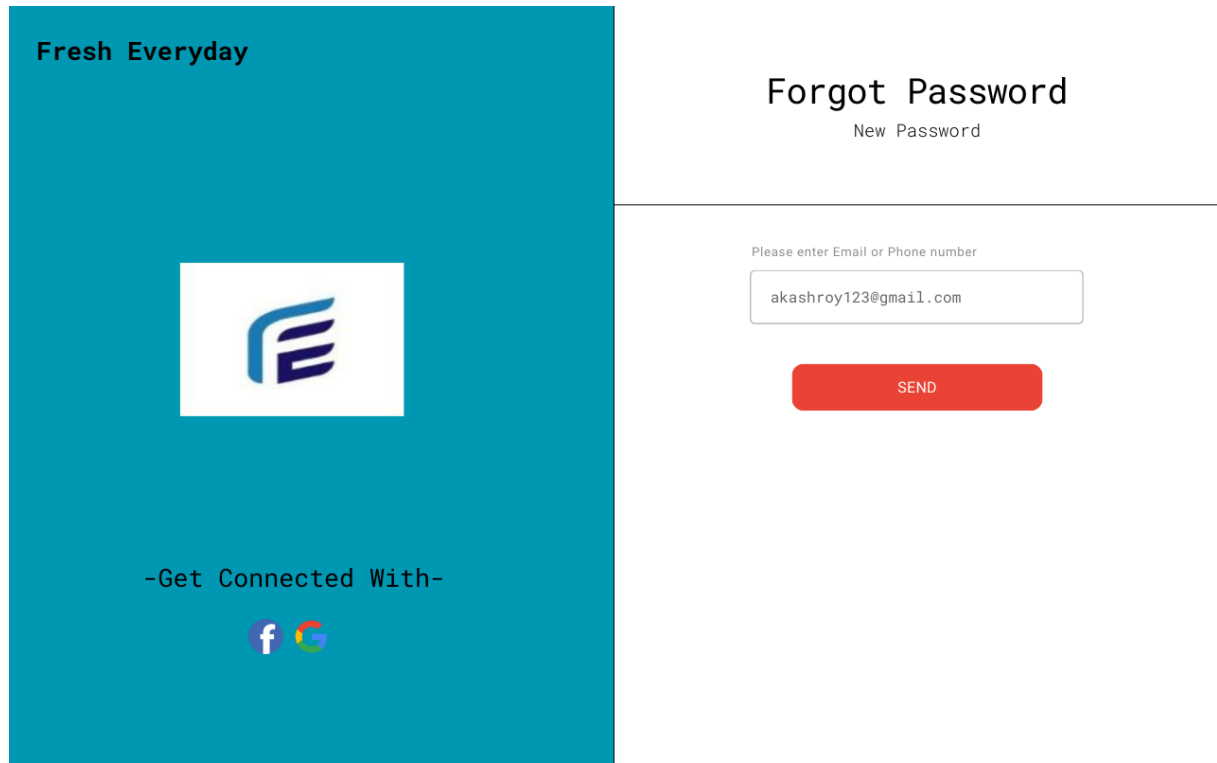


Fig-Forgot Password Page

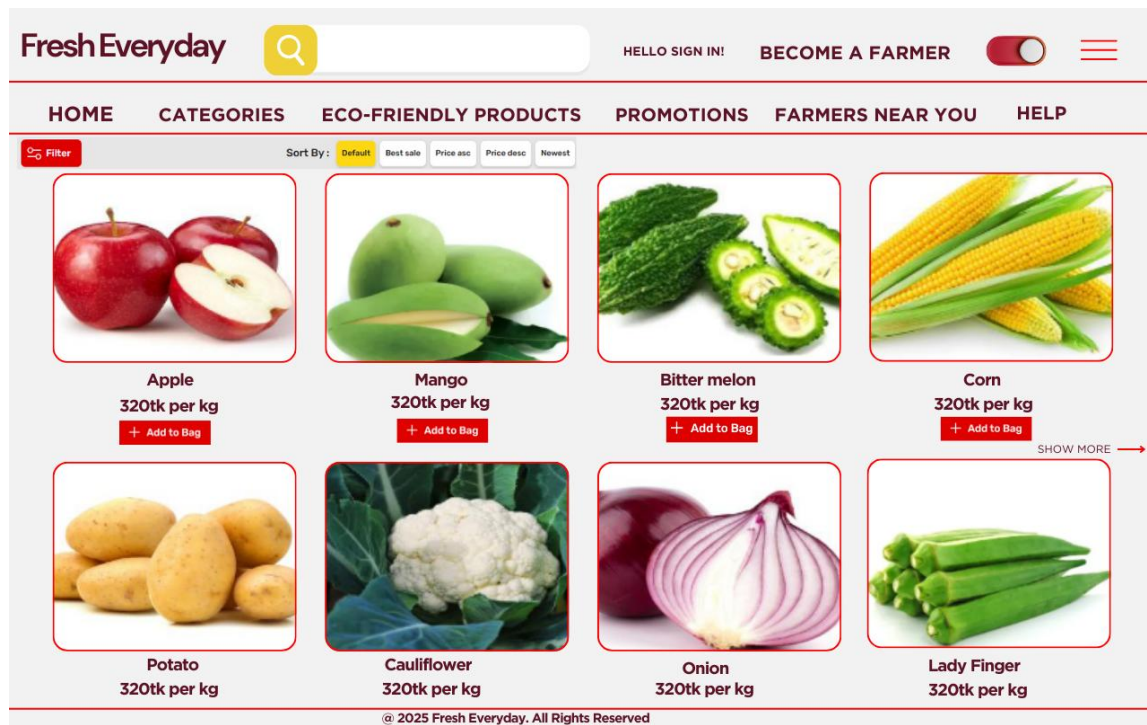


Fig-Home Page



Home

Shop

Product

About



New Product

Apples

Grown naturally in local farms, these apples are free from harmful chemicals and pesticides. Perfect for a health snack or a delicious recipe ingredient.

Farmer : Akash Roy

Location : Kuril bisshoroad, Dhaka



Certified Organic



20 Ratings | 14 Reviews ▾



In Stock (14 Kg Available)



400 TK.
Per kg

♥ Add to Favourite List

🛒 Add to cart

Fig- Product Details Page

FreshEveryday

HOME

SHOP

PRODCT

CART



Sunday best deals!



SHOW MORE →



Fresh Vegetables



Fresh Fruits



Dry Fruits



Dry Vegetables

Fig- Product Deals Page

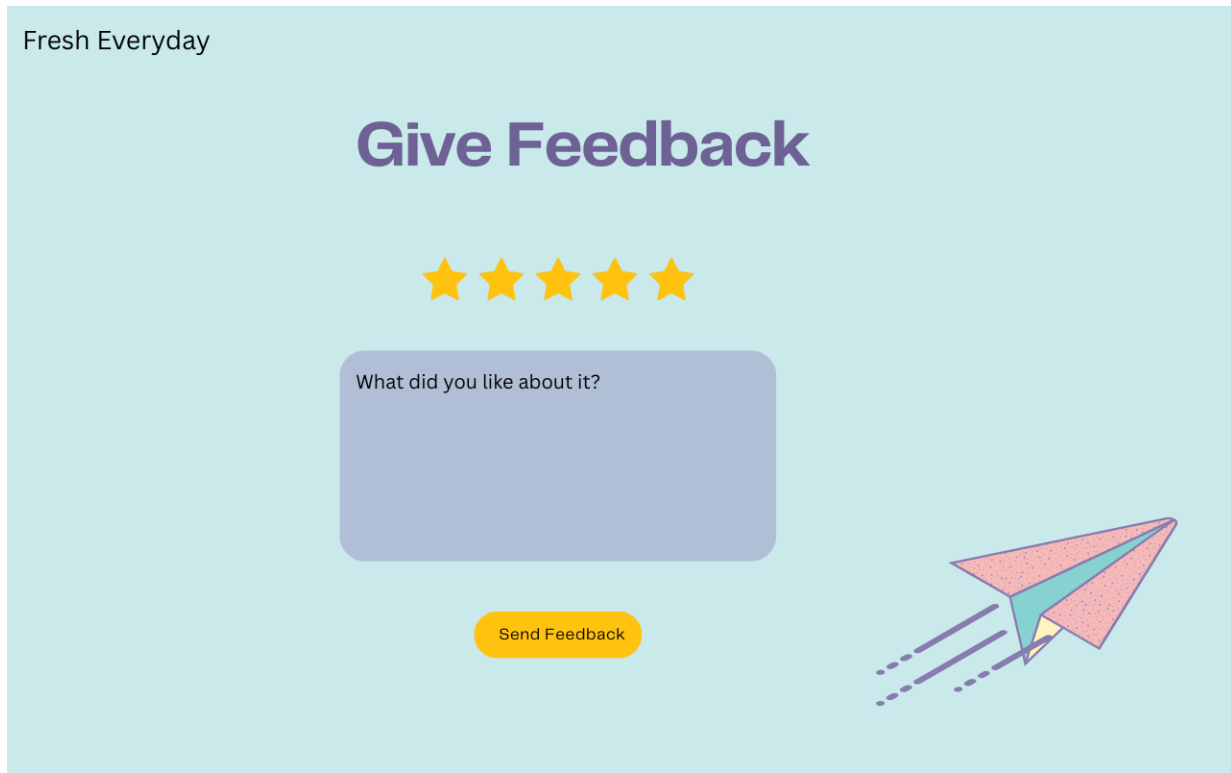


Fig- Feedback Page

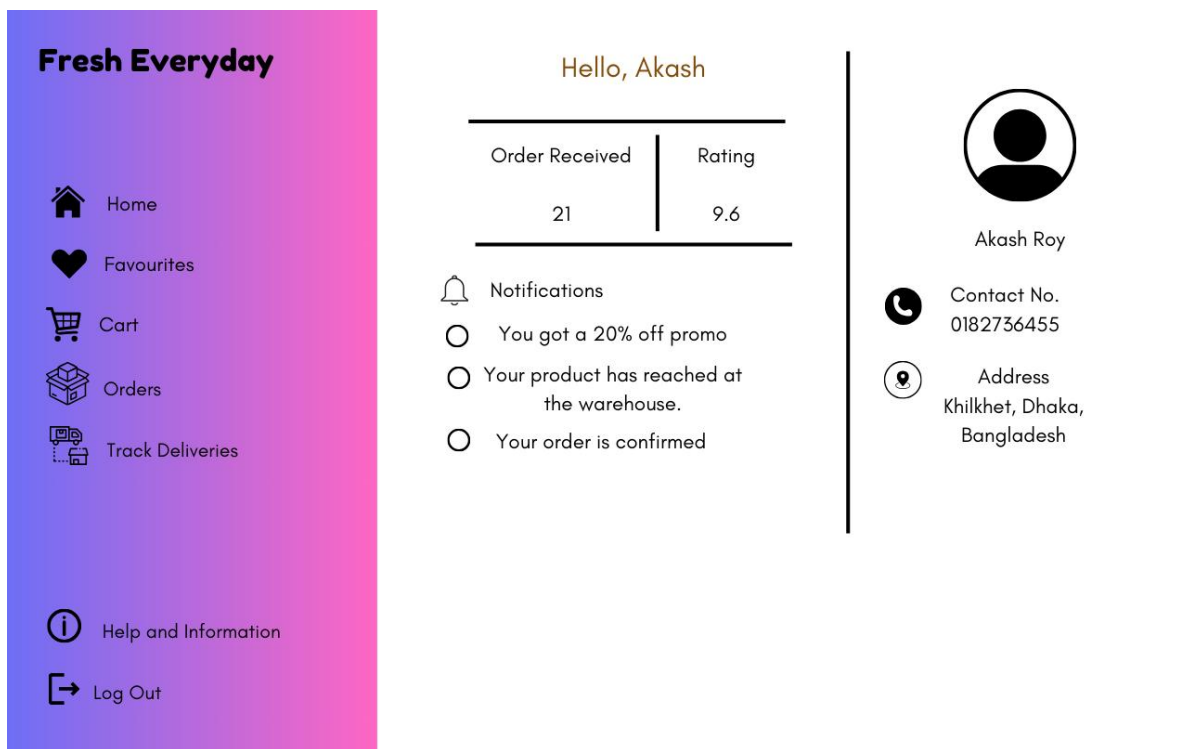
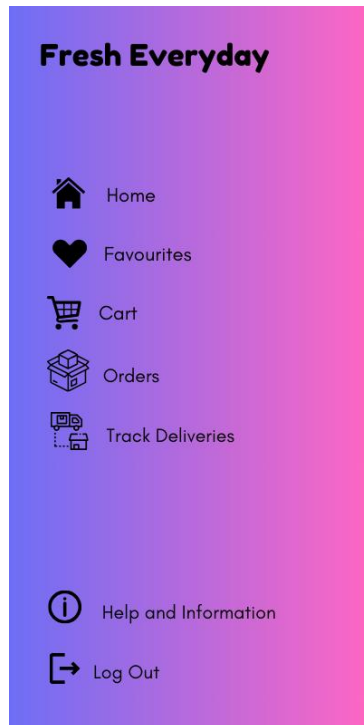


Fig- Consumer Dashboard Page



Profile Settings



[Change Picture](#)
[Delete Picture](#)

First Name*

Last Name*

Contact No.*

Address*

Eco-Friendly Practices

Certifications

No File Chosen

[Submit Profile](#)

Fig- Farmer Profile Setup Page


Fresh Everyday


Consumer Info


Favourites


Notification


Setting


Log out



Akash Roy
 Khilkhet, Dhaka

First Name*

Last Name*

Contact No*

Postal Code*

City*

Apartment

Address*

Date Of Birth

[Save Changes](#)

Fig- Consumer Info Page

Fresh Everyday

HOME PRODUCT ABOUT CONTACT

Hello!



UPLOAD PROFILE

Akash Roy

Update Information Consumer

Email

Phone Number

Address

Shipping Preference

Update Back

Fig- Update Consumer Info Page

Fresh Everyday

Hello!



UPLOAD PROFILE

AKASH FARM

AKASHROY654

Update Information Farmer

Email

Phone Number

Address

Farm Name

Crop Typees

Certifications

Update Back

Fig- Update Farmer Info Page

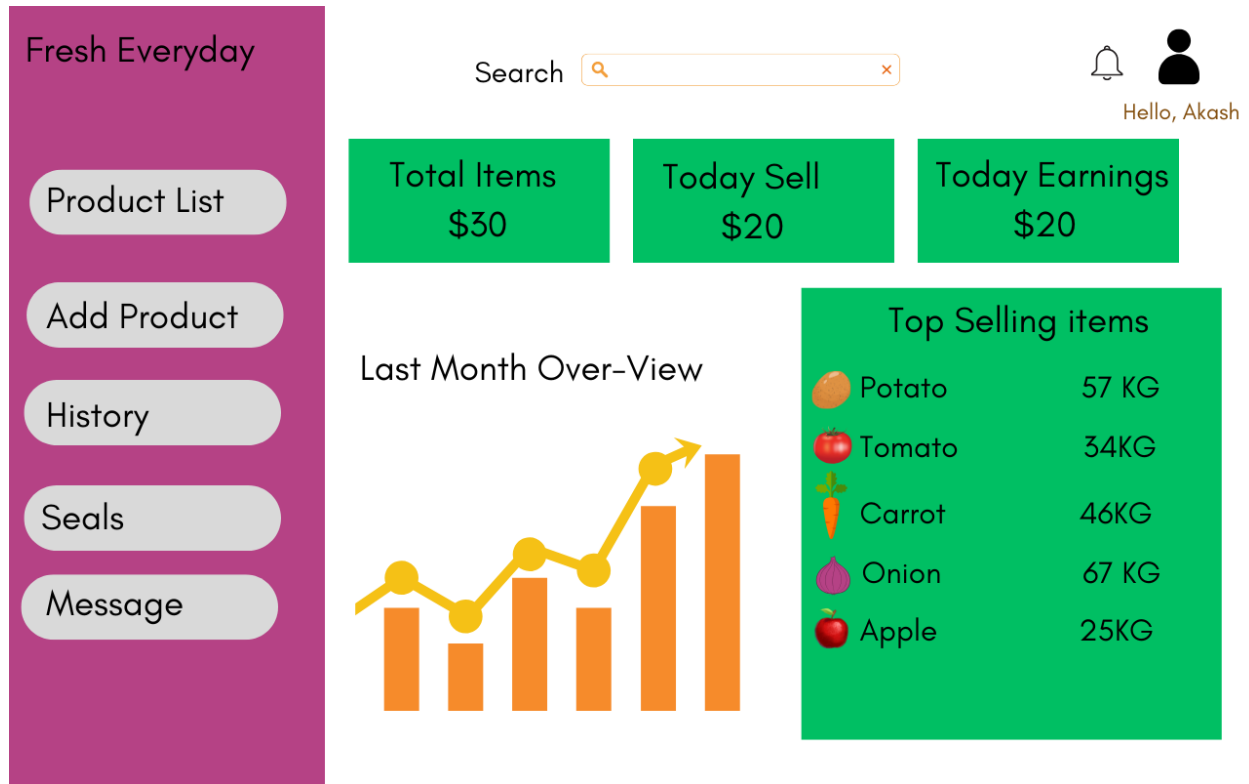


Fig- Farmer Dashboard

The Add Product Page features a purple sidebar with the following navigation options: Add Product, Delete Product, and Add Offers. The main content area includes a search bar, a user profile with a bell icon and the text "Hello, Akash", and a large gray placeholder for a product image. Below the image is a "Select Picture" button. The product form consists of four rows, each with a label and a text input field: "Product Name", "Total Weight(KG)", "Product Price", and "Product Description".

Fig- Add Product Page

Fresh Everyday

Add Product

Delete Product

Add Offers

Search

Select Picture

Product Name

%

Terms and Conditions

Hello, Akash

Fig- Add Offers Page

Fresh Everyday				
Home Shop Product About				
Your Cart				
	Item	Price	Quantity	Total
	Apple	40.00 TK	- 3 +	120.00 TK
	Potato	40.00 TK	- 2 +	80.00 TK
	Cauliflower	40.00 TK	- 5 +	200.00 TK
Subtotal				400.00 TK
Check Out				

Fig- Cart Interface

Transition History & Order History

Date	Type	Amount	Status	Receipts
02-01-2025	Purchase	50.00 Tk.	Complete	View Receipt
02-01-2025	Purchase	50.00 Tk.	Complete	View Receipt
02-01-2025	Purchase	50.00 Tk.	Complete	View Receipt

Order ID	Date	Amount	Status	Receipts
1345	02-01-2025	50.00 Tk.	Delivered	View Receipt
4567	02-01-2025	50.00 Tk.	Delivered	View Receipt
4523	02-01-2025	50.00 Tk.	Delivered	View Receipt

[Back](#)

Fig- Transition and Order History

Checkout

Shopping Address

First Name*	Last Name*
Contact No*	Postal Code*
City*	Floor No.
Address*	Apartment

[Place Order](#)

Order Summary

Subtotal(3)	550.00 Tk.
Delivery Charge	60.00 Tk.
Discount	×
Grand Total	610.00 Tk

Apply Promo Code

[Apply](#)

Fig- Checkout Page

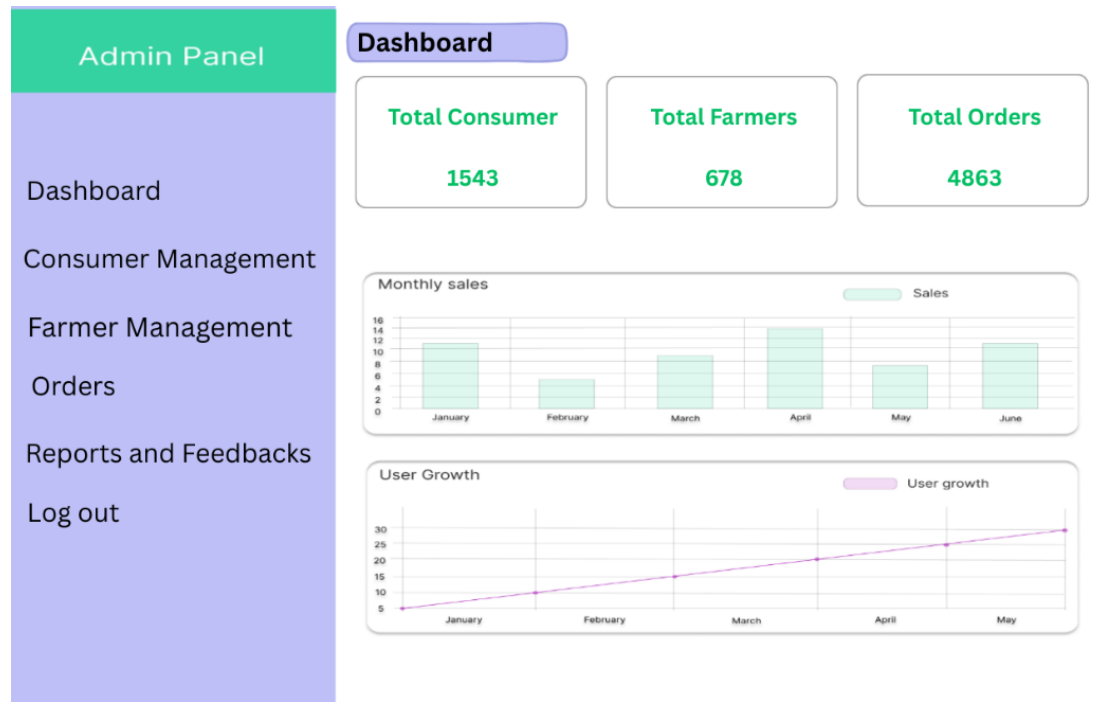


Fig- Admin Dashboard

Admin Panel

Dashboard

Consumer Management

Farmer Management

Orders

Reports and Feedbacks

Log out

Consumer Management

Search

Consumer ID	Name	Email	Registered on	Status	Action
AB01	Akash Islam	ak3@gmail.com	7-1-2025	Active	Edit Delete
AB34	Ashik Islam		7-7-2025	inactive	Edit Delete

«

1

2

3

»

Fig- Consumer Management page

Admin Panel

Dashboard
Consumer Management
Farmer Management
Orders
Reports and Feedbacks
Log out

Farmer Management

Search

Farmer ID	Name	Location	Registered on	Status	Action
YG01	Ibrahim Ullah	Cumilla	7-1-2025	Active	Edit Delete
YG34	Ashik Islam	Noakhali	7-7-2025	inactive	Edit Delete

<<
1
2
3
>>

Fig- Farmer Management page

Fresh Everyday

Dashboard
Companies
All Workers
New Request
Companies Request
Worker Request

Notification

Akash Roy
Invite your farmer platform
3:34pm
07-01-2025

New Team Member Added
3:54pm
07-01-2025

Insufficient Product Alert
3:34pm
09-01-2025

Fig- Notification page

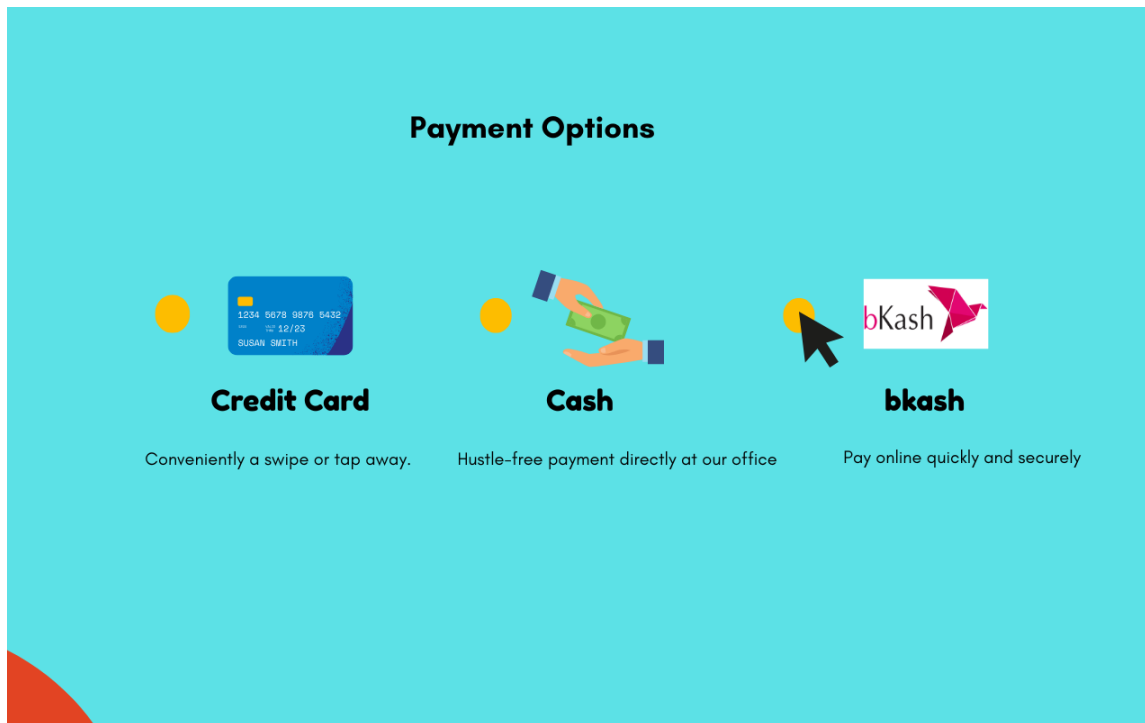


Fig- Payment page

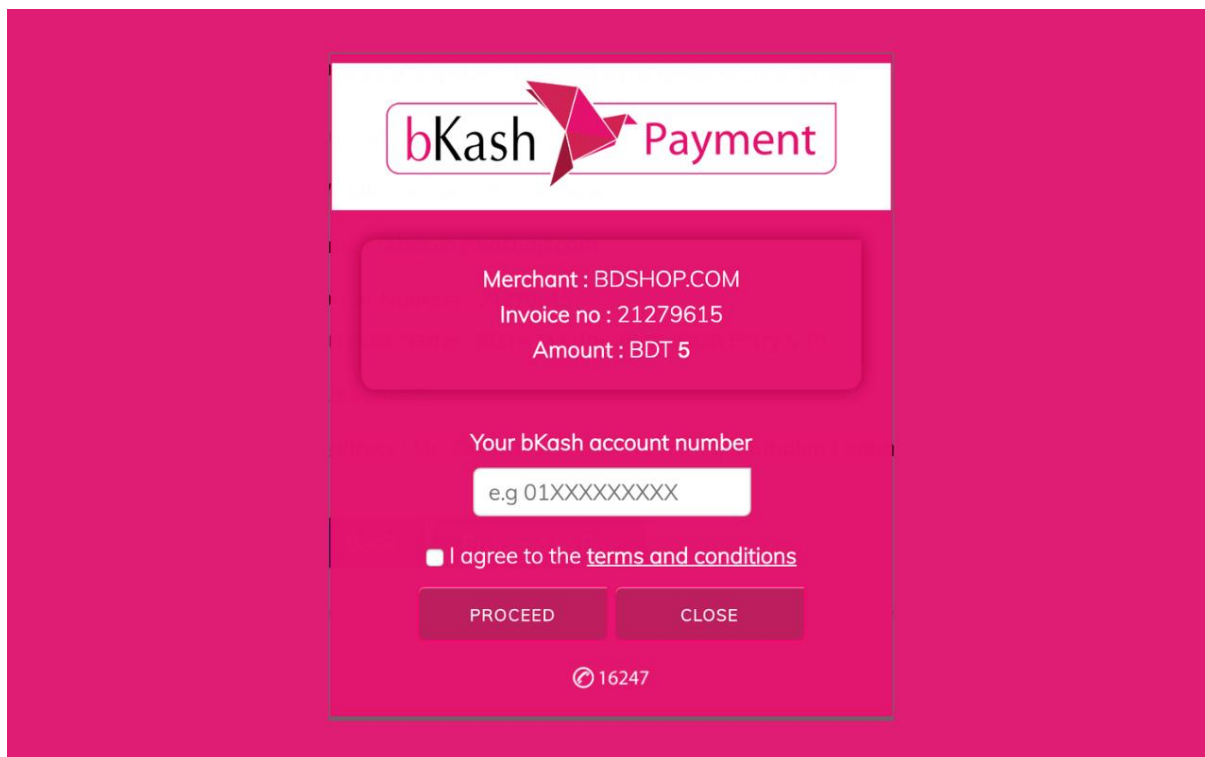


Fig- Transition page

Fresh Everyday

TRACK ORDER



Order Number

Email

Track

- ✓  Ordered
- ✓  Order Ready
- ✓  Shipped
- ✓  Out for Delivery
- ✓  Delivered

Fig- Track Order page

2. Test Plan and Test Cases

TEST CASE-01

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_1		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Sign Up Session		Test Execution date:		
Test title: Verify Consumer Sign Up page with consumer details				
Description: Test Sign Up page				
Precondition (if any): The User must open the Sign-Up page				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Click Sign Up as a consumer 3. Enter All Details 4. Click submit	Name: Akash Roy Email: akashroy1@gmail.com Phone: 01709801663 Address: Khilkhet, Dhaka	User should see a pop-up message that says Sign Up Successful.		
Post Condition:				

TEST CASE-02

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_2		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Sign Up Session		Test Execution date:		
Test title: Verify Farmer Sign Up page with farmer details				
Description: Test Sign Up page				
Precondition (if any): The User must open the Sign-Up page				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
5. Go to the website 6. Click Sign Up as a Farmer 7. Enter All Details 8. Click submit	Name: Roy Email: roy2@gmail.com Phone: 017098016643 Address: Nikunjo, Dhaka	User should see a pop-up message that says Sign Up Successful.		
Post Condition:				

TEST CASE-03

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_3		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Sign Up Session		Test Execution date:		
Test title: Verify Admin Sign Up page with admin details				
Description: Test Sign Up page				
Precondition (if any): The User must open the Sign-Up page				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
9. Go to the website 10. Click Sign Up as a admin 11. Enter All Details 12. Click submit	Name: Afridi Email: afridi10@gmail.com Phone: 01709801663 Address: Kuril, Dhaka	User should see a pop-up message that says Sign Up Successful.		
Post Condition:				

TEST CASE-04

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_4		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Login Session		Test Execution date:		
Test title: Verify login with username and password				
Description: Test the website login page				
Precondition (if any) : User must have valid username and password				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Enter Username 3. Enter password 4. Click submit	Username: 999999999999 Password: 321	User should login to the application		
Post Condition:				

TEST CASE-05

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_5		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed by:		
Module Name: Payment Gateway		Test Execution date:		
Test title: Verify payment System				
Description: Test website payment page and options				
Precondition (if any) : User must login to the system and must select a product to buy				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Enter Username 3. Enter password 4. Click submit 5. Choose a product 6. Click buy button	Method: Bkash Number: 01709801663 OTP: random	User should see a popup message that says payment successful and redirect to home page		
Post Condition:				

TEST CASE-06

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_6		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed by:		
Module Name: Review and Rating		Test Execution date:		
Test title: Verify review and rating system				
Description: Check review and rating system works and update real time				
Precondition (if any) : User must login to this account				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Enter Username 3. Enter password 4. Click submit 5. Click review and rating	Review : good product Rating: 5 star	User should see his comment and rating real time		
Post Condition:				

TEST CASE-07

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_7		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Farmer Dashboard		Test Execution date:		
Test title: Verify Product listing functionality				
Description: Test if farmers can list a product successfully				
Precondition (if any) : Farmer must be logged in				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Login as Farmer 3. Navigate to the “Add-new-product” Section 4. Enter Product Details 5. Click Submit	Name: Tomato Price: \$2/kg Quantity:100kg	Farmer should be able to save the product details also product should appear in the product listing and inventory count should be updated		
Post Condition:				

TEST CASE-08

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_8		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Farmer Dashboard		Test Execution date:		
Test title: Verify inventory management functionality				
Description: Test to see if farmers can update inventory for their products				
Precondition (if any) : Farmer must have listed products				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the “Inventory Management” section 2. Update product inventory 3. Save	Product: Tomato New Quantity:80kg Updated Price:3\$/kg	Update Inventory should be saved and notifications should be sent if necessary		
Post Condition:				

TEST CASE-09

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_9		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Consumer Home Page		Test Execution date:		
Test title: Verify product search functionality				
Description: Test to see if consumers can search for products using filters				
Precondition (if any) : Consumer must be logged in				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Login as Consumer 2. Navigate to the “Search” section 3. Enter criteria section 4. Apply filters(category,price,location,etc.)	Category: Vegetables Price Range: \$1- \$10 Location: Local	Relevant products should be displayed based on search criteria The product list should update dynamically based on selected filters		
Post Condition:				

TEST CASE-10

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_10		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Admin Dashboard		Test Execution date:		
Test title: Verify admins ability to manage user accounts				
Description: Test to see if the admin can view, suspend or delete user accounts				
Precondition (if any) : Admin must have login				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the admin dashboard 2. Navigate to the user management section 3. view a user account 4. Delete a user account	Username: admin Password: admin123 User ID:12345 User ID:12346	User Account with ID 12346 should be deleted from the database		
Post Condition:				

TEST CASE-11

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_11		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Payment Session		Test Execution date:		
Test title: Verify accurate calculation of delivery cost				
Description: Test if the delivery cost is calculated correctly based on distance				
Precondition (if any) : Delivery distance data should be available				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Login as Consumer 2. Add a product to the cart 3. Proceed to checkout	Product: Tomatoes Quantity: 5kg Delivery Address: Khilkhet, Dhaka (15km from farm)	Delivery cost should be calculated based on the 15km distance and displayed correctly in the order summary		
Post Condition:				

TEST CASE-12

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_12		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed by:		
Module Name: Payment Session		Test Execution date:		
Test title: Verify subscription-based delivery option				
Description: Test to see if consumers can subscribe to a monthly delivery service				
Precondition (if any): Consumer must be logged in and have payment details saved				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Login as Consumer 2. Navigate to the subscription option . 3. Select a monthly delivery plan and make payment	Username: Consumer1 Password: 12345 Plan: \$20/month Payment: Credit Card/Mobile banking options	Consumers should be able to select the monthly delivery plan and complete the payment successfully. A confirmation pop-up message should appear indicating the payment was successful and be directed to the homepage		
Post Condition:				

TEST CASE-13

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_13		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed by:		
Module Name: Consumer Dashboard		Test Execution date:		
Test title: Verify the ability to add and remove products from the favourite list				
Description: Test the consumer’s ability to manage their favourite list				
Precondition (if any) : Consumer must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Navigate to the product catalog. 2. Select a product and add it to the favourite list.	Product: Tomatoes	Product is added to the product list and removed from the favourite list without errors.		
3. Navigate to the Favourite list and verify the product is added. 4. Remove the product from the favourite list. 5. Verify the product is removed successfully from the favourite list.				
Post Condition:				

TEST CASE-14

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_14		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Payment Session		Test Execution date:		
Test title: Verify accurate delivery cost calculation for subscription orders				
Description: Test the system’s ability to calculate delivery costs accurately for monthly subscription orders.				
Precondition (if any) : Consumer must be logged in and subscribed to a monthly delivery plan.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in as a consumer. 2. Add a product to the cart. 3. Proceed to checkout. 4. Verify that the delivery cost is calculated based on the subscription plan.	Product: Apples Subscription Plan: \$20/month	Delivery cost is waived or discounted as per the subscription plan.		

Post Condition:

TEST CASE-15

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_15		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed by:		
Module Name: Farmer Dashboard		Test Execution date:		
Test title: Verify farmers receive inventory alerts for low stock				
Description: Test whether farmers are notified when inventory levels fall below a threshold.				
Precondition (if any) : Farmer must have products listed in their inventory.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in as a farmer. 2. Navigate to the inventory management section. 3. Reduce the stock of a product below the set threshold. 4. Verify that a notification or alert is sent to the farmer.	Product: Potatoes Low Stock Threshold: 10kg	Farmer receives an inventory alert when the stock level is below 10kg.		
Post Condition:				

TEST CASE-16

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_16		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Sign Up Session		Test Execution date:		
Test title: Verify a Prevent Duplicate Email/Phone in Sign-Up				
Description: Test when farmer/consumer trying to sign up with an already registered email or phone number				
Precondition (if any) : The user must open the Sign-Up page				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Click Sign Up as a consumer/farmer 3. Enter a duplicate email and phone number 4. Click submit	Name: Akash Roy Email: akash48054@gmail.com Phone: 01700000000 Address: Khilkhet,Dhaka	System should show 'Email or Phone number already exists' message.		
Post Condition:				

TEST CASE-17

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_17		Test Designed date:		
Test Priority (Low, Medium, High): High Module Name: Sign Up Session		Test Executed by: Test Execution date:		
Test title: Verify a Reject Weak Password During Sign-Up				
Description: Test a weak passwords are rejected during sign-up				
Precondition (if any) : The user must open the Sign-Up page.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Click Sign Up as a consumer 3. Enter weak password 4. Click submit	Name: Akash Roy Password: 12345 Email: akash48054@gmail.com Phone: 01700000001 Address: Khilkheta, Dhaka	System should reject weak password and prompt user.		
Post Condition:				

TEST CASE-18

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_18		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Sign Up Session		Test Execution date:		
Test title: Verify an OTP Verification During Sign-Up				
Description: Test OTP verification is required and successful				
Precondition (if any) : User must enter a valid sign-up info				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to the website 2. Click Sign Up 3. Submit the sign-up form 4. Receive OTP 5. Enter OTP 6. Click verify	Name: Akash Roy Password: akash49com Email: user18@gmail.com Phone: 01700000002 Address: Khilkheta, Dhaka.	User account should be created after successful OTP verification.		
Post Condition:				

TEST CASE-19

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_19		Test Designed date:		
Test Priority (Low, Medium, High): High Module Name: Login Session		Test Executed by: Test Execution date:		
Test title: Verify a Google or Facebook Sign-In Integration				
Description: Test whether Google authentication works properly				
Precondition (if any) : User must clicks on 'Sign in with Google' button				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Click 'Sign in with Google' 2. Select Google account 3. Give permissions 4. Go to the dashboard	Google Account: akash48054@gmail.com	User should be redirected to dashboard after Google login.		
Post Condition:				

TEST CASE-20

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_20		Test Designed date:		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Login Session		Test Execution date:		
Test title: Verify a block Login After 3 wrong password Failed Attempts				
Description: Ensure account is blocked after 3 incorrect login attempts for the next 30 minutes.				
Precondition (if any) : User enters wrong password				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Go to website 2. enter username and password. 3. Attempt login with wrong password 3 times 4. Check error message	Username : Akash Roy Wrong Password: abc123	System should block user for 30 minutes with a message.		
Post Condition:				

TEST CASE-21

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_21		Test Designed date:		
Test Priority (Low, Medium, High): Medium Module Name: Farmer Dashboard		Test Executed by: Test Execution date:		
Test title: Verify a Monthly Sales Graph Display				
Description: Test to Ensure graph loads with correct data				
Precondition (if any) : Farmer has previous sales data				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1.Go to website 2.Log in as Farmer 3. Go to Farmer dashboard 4. Check sales graph section	Sales records for last 3 months	Monthly sales graph should display properly		
Post Condition:				

TEST CASE-22

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_22		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed by:		
Module Name: Consumer Dashboard		Test Execution date:		
Test title: Verify a Notification List				
Description: Test notifications are shown on consumer dashboard				
Precondition (if any) : Consumer must logged in				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1.Go to website 2.Login as Consumer 3. Click on notification bell icon 4. Check list of notifications	Notifications: Offers, Order updates	All recent notifications should be visible.		
Post Condition:				

TEST CASE-23

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Akash Roy		
Test case ID: FR_23		Test Designed date:		
Test Priority (Low, Medium, High): Medium		Test Executed by:		
Module Name: Payment Page		Test Execution date:		
Test title: Verify payment failure handling in bKash option				
Description: Test system behavior when bKash payment fails				
Precondition (if any) : User must be on the payment page with items in cart				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1.Go to website 2.Login as Consumer 3. select item to the cart 4.Select bKash as payment method 5.Enter invalid mobile number or pin 6.Click Pay button	Mobile Number: 01827364536 Wrong Pin: 0000	System should show payment failure message		
Post Condition:				

TEST CASE-24

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Abdullah Al Taieb		
Test case ID: FR_24		Test Designed date:		
Test Priority (Low, Medium, High): Low		Test Executed by:		
Module Name: Navigation bar		Test Execution date:		
Test title: Check all navigation menus are displaying properly.				
Description: Getting navigation menu without any error.				
Precondition (if any): The user must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the website. 2. Look at the top for menus. 3. For mobile device click hamburger to check if the menus are there.	User id, Password	Users should see a navigation bar with some menus.		
Post Condition:				

TEST CASE-25

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Abdullah Al Taieb		
Test case ID: FR_25		Test Designed date:		
Test Priority (Low, Medium, High): Low		Test Executed by:		
Module Name: Navigation bar		Test Execution date:		
Test title: Check all paginations are working properly.				
Description: Changing pages from navigation menu to different pages .				
Precondition (if any): The user must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the website. 2. Look at the top for menus. 3. For mobile device click hamburger to check if the menus are there. 4. Click your desired page menu.	User id, password, home page, product page, signup page, sign in page, cart page, payment page, notifications and account page.	Users should go to the clicked page.		
Post Condition:				

TEST CASE-26

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Abdullah Al Taieb		
Test case ID: FR_26		Test Designed date:		
Test Priority (Low, Medium, High): Low		Test Executed by:		
Module Name: Profile page		Test Execution date:		
Test title: Check all items are displaying properly.				
Description: Going to the profile page and checking all items displayed properly.				
Precondition (if any): The user must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the website. 2. Look at the top for menus. 3. For mobile device click hamburger to check if the menus are there. 4. Click the circular photo icon from the top right corner.	User id, password, navigation menu, profile page icon	Users should see a profile page with all the functionalities.		
Post Condition:				

TEST CASE-27

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Abdullah Al Taieb		
Test case ID: FR_27		Test Designed date:		
Test Priority (Low, Medium, High): Low		Test Executed by:		
Module Name: Profile page		Test Execution date:		
Test title: Add photo functionality is working properly.				
Description: Going to the profile page and uploading a new photo.				
Precondition (if any): The user must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the website. 2. Look at the top for menus. 3. For mobile device click hamburger to check if the menus are there. 4. Click the circular photo icon from the top right corner. 5. Click upload profile picture. 6. Select picture from device to upload. 7. Click confirm.	User id, password, navigation menu, profile page icon, profile page, upload profile picture.	Users should see a profile picture is being displayed in the picture box and also on profile page icon.		
Post Condition:				

TEST CASE-28

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Abdullah Al Taieb		
Test case ID: FR_28		Test Designed date:		
Test Priority (Low, Medium, High): Low		Test Executed by:		
Module Name: Profile page		Test Execution date:		
Test title: Change photo functionality is working properly.				
Description: Going to the profile page and changing the previous photo.				
Precondition (if any): The user must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the website. 2. Look at the top for menus. 3. For mobile device click hamburger to check if the menus are there. 4. Click the circular photo icon from the top right corner. 5. Click change profile picture. 6. Select picture from device to upload. 7. Click confirm.	User id, password, navigation menu, profile page icon, profile page, change profile picture.	Users should see the new profile picture being displayed in the picture box and also on profile page icon.		
Post Condition:				

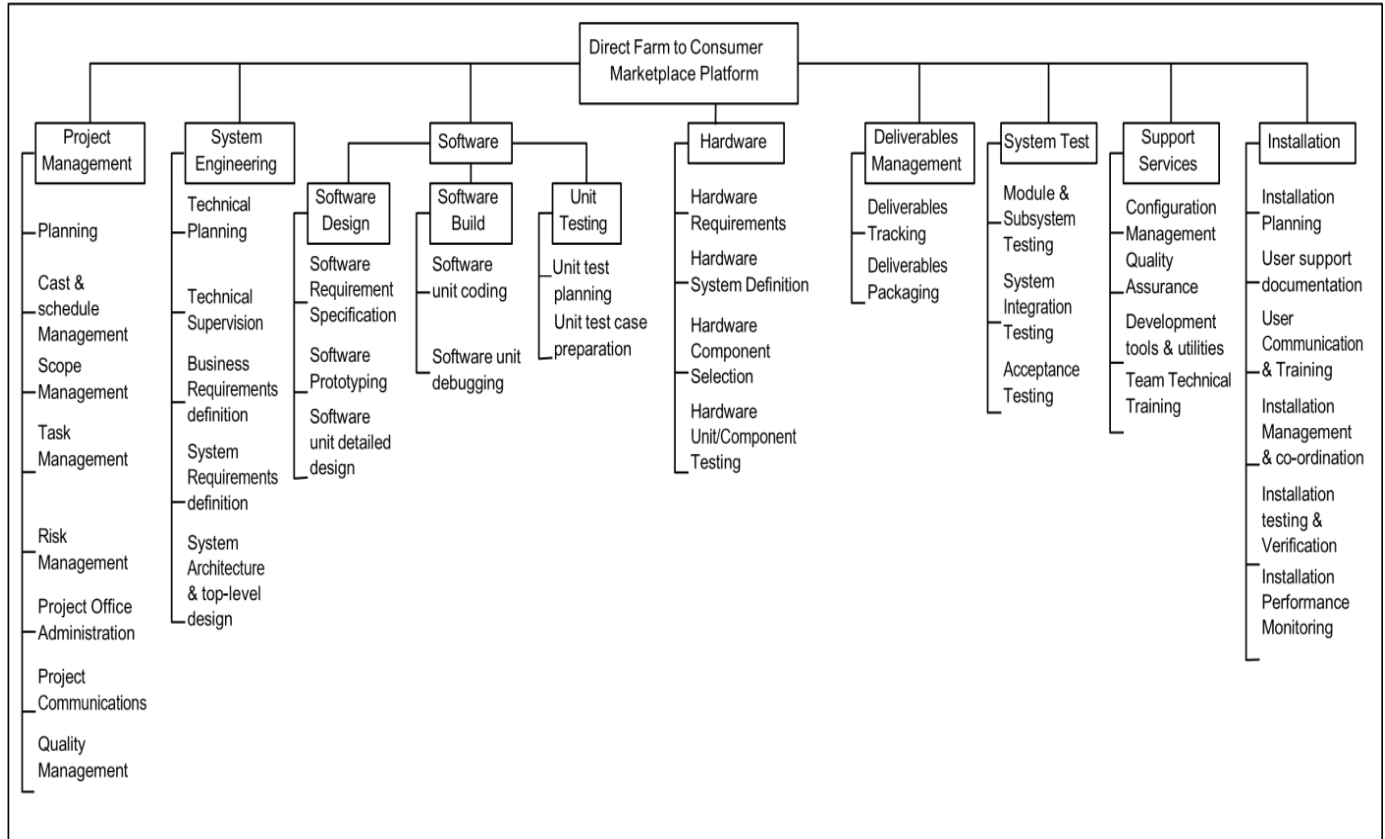
TEST CASE-29

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Abdullah Al Taieb		
Test case ID: FR_29		Test Designed date:		
Test Priority (Low, Medium, High): Low		Test Executed by:		
Module Name: Profile page		Test Execution date:		
Test title: Edit details functionality is working properly.				
Description: Going to the profile page and changing information.				
Precondition (if any): The user must be logged in.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the website. 2. Look at the top for menus. 3. For mobile device click hamburger to check if the menus are there. 4. Click edit [name or address or contact] 5. Click confirm.	User id, password, navigation menu, profile page icon, profile page, edit name, edit address, edit contact.	Users should be able to change his name, address and contact number.		
Post Condition:				

TEST CASE-30

Project name: DIRECT FARM TO CONSUMER MARKETPLACE PLATFORM		Test Designed by: Abdullah Al Taieb		
Test case ID: FR_30		Test Designed date:		
Test Priority (Low, Medium, High): Low		Test Executed by:		
Module Name: Consumer dashboard		Test Execution date:		
Test title: Checking language toggle button.				
Description: Checking if the language of the page changes by clicking the toggle button.				
Precondition (if any): The user must be logged in as consumer.				
Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in to the website. 2. Look at the top for menus. 3. Click language toggle button.	User id, password, language toggle button, English to Bengali and vice versa.	Users should be able to change the language of the website.		
Post Condition:				

3. Work Breakdown Structure



4. Estimation using COCOMO

Assuming a semidetached type of project with SLOC = 6000, Coefficient 3.0, Project complexity(P) 1.12 and SLOC-dependent coefficient (T) = 0.35

$$\text{Effort} = \text{PM} = \text{Coefficient} \times (\text{Effort Factor})^P \times (\text{SLOC}/1000)^T$$

$$= 3.0 * (6000/1000)^{1.12}$$

$$= 22.32$$

$$\text{Development time} = \text{DM} = 2.50 * (\text{PM})^T$$

$$= 2.50 * 22.32^{0.35}$$

$$= 7.41$$

$$\approx 7$$

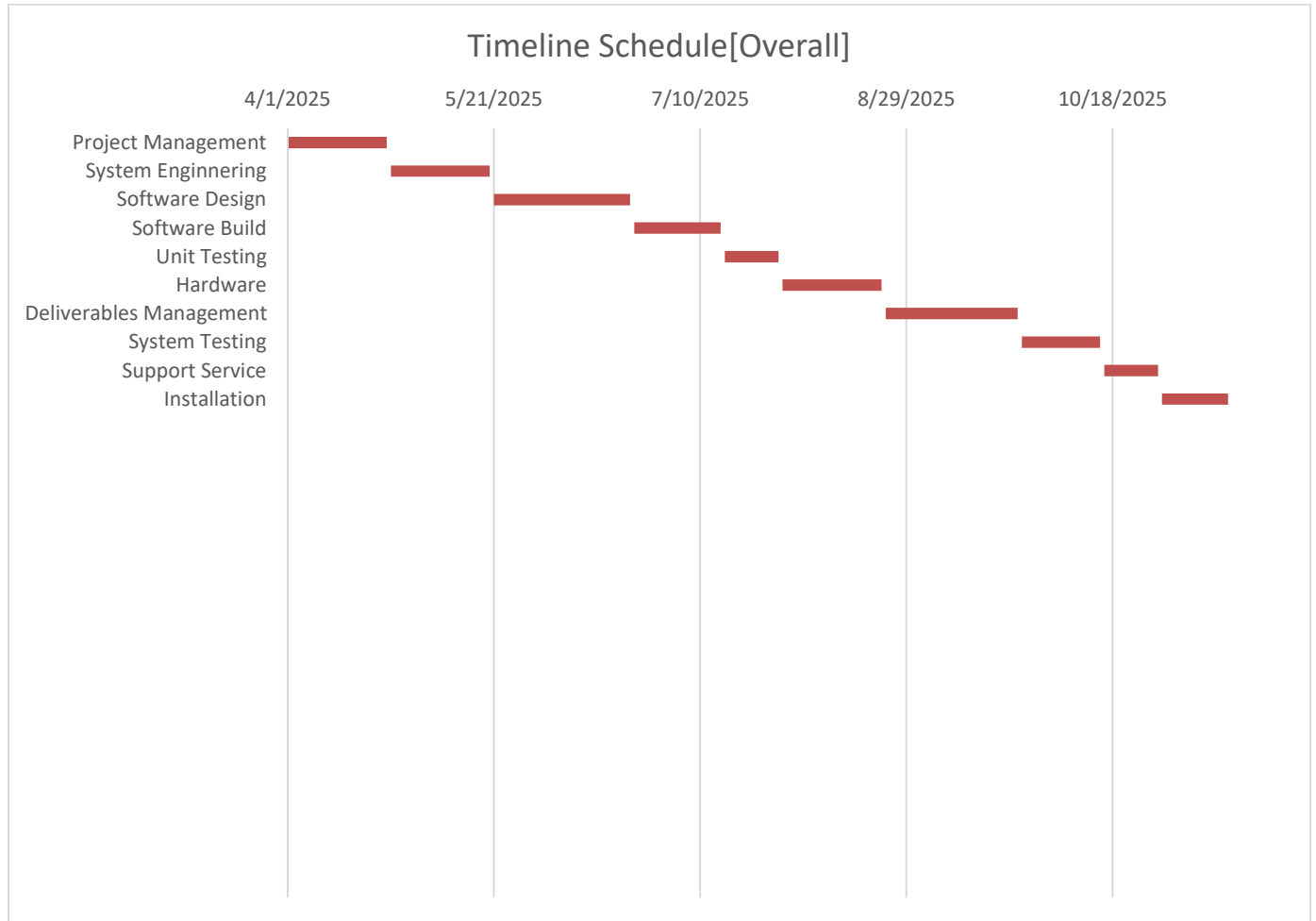
$$\text{Required number of people} = \text{ST} = \text{PM}/\text{DM}$$

$$= 22.32/7.41$$

$$= 3.01$$

$$\approx 3$$

5. Gantt Chart (overall schedule)



Task	Person	Start Date	End Date	Duration
Project Management	Akash	4/1/2025	4/25/2025	24
System Engineering	Akash	4/26/2025	5/20/2025	24
Software Design	Akash	5/21/2025	6/23/2025	33
Software Build	Akash	6/24/2025	7/15/2025	21
Unit Testing	Akash	7/16/2025	7/29/2025	13
Hardware	Taieb	7/30/2025	8/23/2025	24
Deliverables Management	Taieb	8/24/2025	9/25/2025	32
System Testing	Taieb	9/26/2025	10/15/2025	19
Support Service	Taieb	10/16/2025	10/29/2025	13
Installation	Rezaul	10/30/2025	11/15/2025	16

6. Gantt Chart (detailed schedule)



Task	Start Date	End Date	Duration
Planning	4/1/2025	4/3/2025	2
Cost & Schedule Management	4/4/2025	4/5/2025	1
Scope Management	4/6/2025	4/7/2025	1
Task Management	4/8/2025	4/12/2025	4
Risk Management	4/13/2025	4/18/2025	5
Project Office Administration	4/19/2025	4/25/2025	6
Project Communications	4/26/2025	4/30/2025	4

Quality Management	5/1/2025	5/5/2025	4
			0
Technical Planning	5/6/2025	5/9/2025	3
Technical Supervision	5/10/2025	5/12/2025	2
Business Requirements definition	5/16/2025	5/20/2025	4
System Requirements definition	5/21/2025	5/23/2025	2
System Architecture & top-level design	5/24/2025	5/28/2025	4
			0
Software Requirement Specification	5/29/2025	6/2/2025	4
Software Prototyping	6/3/2025	6/6/2025	3
Software unit detailed design	6/7/2025	6/10/2025	3
			0
Software unit coding	6/11/2025	6/17/2025	6
Software unit debugging	6/18/2025	6/21/2025	3
			0
Unit test planning	6/22/2025	6/24/2025	2
Unit test case preparation	6/25/2025	6/29/2025	4
			0
Hardware Requirements	6/30/2025	7/4/2025	4
Hardware System Definition	7/5/2025	7/10/2025	5
Hardware Component Selection	7/11/2025	7/15/2025	4
Hardware Unit/Component Testing	7/16/2025	7/22/2025	6
			0
Deliverables Tracking	7/23/2025	7/26/2025	3
Deliverables Packaging	7/27/2025	8/2/2025	6
			0
Module & Subsystem Testing	8/3/2025	8/12/2025	9
System Integration Testing	8/13/2025	8/23/2025	10
Acceptance Testing	8/24/2025	9/5/2025	12
			0
Configuration Management	9/6/2025	9/10/2025	4
Quality Assurance	9/11/2025	9/16/2025	5
Development tools & utilities	9/22/2025	9/28/2025	6
Team Technical Training	9/29/2025	10/8/2025	9
			0
Installation Planning	10/9/2025	10/15/2025	6
User support documentation	10/16/2025	10/23/2025	7
User Communication & Training	10/24/2025	10/26/2025	2
Installation Management & co-ordination	10/27/2025	11/5/2025	9
Installation testing & Verification	11/6/2025	11/11/2025	5
Installation Performance Monitoring	11/12/2025	11/15/2025	3

7. EVA

Task	Planned Effort	Actual Effort
1	13.5	12
2	8	8.7
3	11	8
4	12.5	12
5	12	14.5
6	6.3	6.7
7	10	10
8	8	7
9	5.5	5.2
10	11	9.5
11	13	13.7
12	17	15
13	7.5	7
14	12	12.5
15	10	12
16	4.5	5.5
17	9	8
18	16	14
19	11	11
20	8	8.7
21	5	6
22	13	18.8
23	14	12
24	8.8	8
25	5.2	6
26	5	7
27	15	
28	7	
29	14	
30	11	
31	4.7	
32	8	
33		
34		
35		

36		
37		
38		
39		
Total = 39	Effort Estimated= 470-person day	

BCWP= 256.8

BCWS= 316.5

ACWP= 258.8

BAC= 470

$SPI = BCWP / BCWS = 256.8 / 316.5 = 0.8113744076$

$SV = BCWP - BCWS = 256.8 - 316.5 = -59.7 \sim -59$

$CPI = BCWP / ACWP = 256.8 / 258.8 = 0.9922720247$

$CV = BCWP - ACWP = 256.8 - 258.8 = -2$

$\% \text{ SCHEDULE FOR COMPLETION} = BCWS / BAC = (316.5 / 470) * 100 = 67.34\%$

$\% \text{ COMPLETION} = BCWP / BAC = (256.8 / 470) * 100 = 54.64\%$

8. Risk Management and Mitigation Plan

Risks	Category	Probability	Impact	RMMM
Miscommunication with stakeholders	CU	35%	1	Establish clear communication channels and frequent reviews.
Market demand shifts unexpectedly	BU	50%	2	Conduct market analysis and plan for contingencies.
Vendors fail to meet deadlines	BU	30%	3	Use contractual incentives for on-time delivery.
Difficulty in onboarding new team members	ST	25%	2	Provide proper training and mentoring for new team members.
Team burnout due to workload	ST	55%	2	Ensure proper workload distribution and provide work-life balance support.
Requirements creep during development	PR	75%	2	Freeze requirements after a baseline and manage changes
Insufficient server capacity	PS	40%	3	Perform load testing and plan for scalability.
Insufficient testing of modules	DE	70%	3	Create a comprehensive test plan and conduct early testing.
Tools used are not scalable	TE	25%	1	Evaluate tool scalability and test with expected load scenarios.

Requirements not properly defined	PR	50%	2	Use prototyping and interviews to clarify requirements.
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Impact Values:

- 1- Catastrophic
- 2- Critical
- 3- Marginal
- 4- Negligible

The work product is called a Risk Mitigation, Monitoring, and Management Plan (RMMM)